

PROJECT PROPOSAL

Intelligent College Assistant Chatbot

Design and Implementation Using Core Artificial Intelligence Techniques

Subject	Artificial Intelligence (S5)
Domain	AI — Search, Knowledge, Planning, Learning
Platform	Python + Flask + Web Interface
Submitted To	Project Supervisor
Academic Year	2025 – 2026

1. Abstract

In today's fast-paced academic environment, students often face difficulties in obtaining timely and accurate information regarding admissions, fees, examination schedules, faculty details, and other college-related queries. Manual helpdesk systems are time-consuming, unavailable outside working hours, and prone to human error. This project proposes the design and implementation of an Intelligent College Assistant Chatbot that leverages core Artificial Intelligence techniques to provide students with instant, accurate, and context-aware responses.

The proposed system is modeled as an Intelligent Goal-Based Agent that perceives user queries as inputs and produces appropriate responses as actions. The chatbot employs uninformed and informed search algorithms including BFS and A* to efficiently navigate the knowledge base. Knowledge is represented using First-Order Logic with Forward and Backward Chaining for inference. A response planning module handles multi-step queries using classical and hierarchical planning. A Decision Tree classifier and Reinforcement Learning mechanism enable the system to improve over time, while an Expert System rule base ensures reliable responses.

This project demonstrates the practical application of AI in education and customer service, covering all six modules of the S5 Artificial Intelligence syllabus in a single, cohesive, and deployable system.

2. Problem Statement

College students frequently require information about a wide range of topics including admission procedures, fee structures, examination schedules, faculty contacts, hostel eligibility, library access, and upcoming events. The current system relies on manual helpdesk operations that suffer from several limitations:

- Help desk staff are unavailable outside working hours, leaving students without support in the evenings, weekends, and holidays.
- Manually searching through notice boards, websites, or waiting in queues is time-consuming and frustrating.
- Different staff members may provide inconsistent or incorrect information, leading to confusion.
- Staff spend significant time repeatedly answering the same common questions, reducing productivity.
- There is no centralised, intelligent system that can handle complex, multi-part queries.

These challenges call for an AI-powered solution that can provide instant, accurate, and 24/7 assistance to students, while also demonstrating the application of core AI techniques studied in the curriculum.

3. Objectives

- **Design and implement an Intelligent College Assistant Chatbot using Python.**
- Model the chatbot as a Goal-Based Intelligent Agent following Module 1 principles.
- Implement BFS and A* search algorithms to navigate the knowledge base efficiently.

- Represent college knowledge using First-Order Logic with Forward and Backward Chaining.
- Incorporate a response planning module using classical and hierarchical planning techniques.
- Implement a Decision Tree classifier for query categorisation and Reinforcement Learning for improvement.
- Build an Expert System rule base as the core reasoning engine.
- Develop a user-friendly web interface for student interaction.
- Demonstrate the real-world application of AI in education and customer service.
- Cover all six modules of the S5 Artificial Intelligence syllabus in a single project.

4. Syllabus Modules Covered

This project uniquely integrates concepts from all six modules of the S5 Artificial Intelligence syllabus, making it a comprehensive demonstration of the course curriculum.

Module	Topic	Application in Chatbot	Key Concepts
Module 1	Intelligent Agent Design	Chatbot modeled as Goal-Based Agent with percepts, environment, and actions	PEAS, Agent Types
Module 2	Search Algorithms	BFS searches KB layer-by-layer; A* finds shortest reasoning path using heuristics	BFS, A*, Heuristics
Module 3	Knowledge Representation	FOL rules store college facts; Forward Chaining derives answers; Backward Chaining verifies eligibility	FOL, Chaining, Resolution
Module 4	Planning	Classical Planning sequences response steps from query receipt to answer delivery	Classical & Hierarchical Planning
Module 5	Learning & Expert Systems	Decision Tree classifies queries; RL updates from feedback; Expert System manages rule base	Decision Tree, RL, Expert Systems
Module 6	AI Applications	Real-world deployment in college for education support and customer service helpdesk	AI in Education, AI in Customer Service

5. Core Algorithms Used

5.1 Breadth First Search (BFS)

BFS explores the knowledge base tree level by level to retrieve answers to simple, direct queries such as fee amounts, exam dates, and contact information. It guarantees finding the closest matching answer first and is used when the query maps directly to a known fact.

5.2 A* Algorithm

The A* algorithm is employed for complex queries that require multi-step reasoning. Using the evaluation function $f(n) = g(n) + h(n)$, where $g(n)$ is the cost of steps taken and $h(n)$ is the heuristic estimate of remaining cost, A* finds the most efficient reasoning path to the correct answer with minimal computation.

5.3 Forward Chaining

Forward Chaining starts from known facts about the user's query and applies IF-THEN production rules iteratively to derive a conclusion. It is used for straightforward queries where sufficient information is provided upfront, such as displaying the fee for a specific course and year.

5.4 Backward Chaining

Backward Chaining starts from a goal state and works backwards to determine whether the available facts support it. It is used for eligibility verification queries such as hostel admission checks and scholarship eligibility, where the system must confirm or deny a condition.

5.5 Decision Tree

A Decision Tree classifier is the first module triggered when a student sends a message. It classifies the incoming query into predefined categories (fees, exams, hostel, admission, faculty, events) and routes it to the appropriate section of the knowledge base and inference engine.

5.6 Reinforcement Learning

The chatbot incorporates a simple Reinforcement Learning mechanism in which student feedback (thumbs up or thumbs down) acts as a reward or penalty signal. Over time, the system learns to prefer response patterns that received positive feedback and avoids those that were marked unhelpful.

5.7 Constraint Satisfaction Problem (CSP)

CSP techniques are applied for scheduling-related queries, such as finding a free time slot for a faculty meeting or resolving timetable conflicts. The system identifies a solution that satisfies all constraints simultaneously, including the student's schedule, faculty availability, and room bookings.

5.8 Expert System with Production Rules

The core reasoning engine of the chatbot is an Expert System comprising a curated set of IF-THEN production rules representing all college knowledge. This ensures reliable, deterministic responses for known queries even in the absence of machine learning training data, and allows easy updates when college information changes.

6. System Architecture

The chatbot is structured as a layered architecture where each layer corresponds to a distinct AI module from the syllabus. The following describes the data flow from user input to final response:

- User Input Layer: The student types a query through the web interface or messaging platform.

- Agent Interface (Module 1): The Intelligent Agent perceives the query as its environment input and determines the appropriate action.
- Query Classification (Module 5): The Decision Tree classifier identifies the category of the query and routes it accordingly.
- Search Engine (Module 2): BFS or A* searches the knowledge base to find relevant facts and reasoning paths.
- Inference Engine (Module 3): Forward or Backward Chaining applies FOL rules to derive or verify the answer.
- Response Planner (Module 4): The planning module sequences the steps required to generate a complete, accurate response.
- Learning Engine (Module 5): Reinforcement Learning updates internal weights based on student feedback after each interaction.
- Output Layer (Module 6): The formatted answer is delivered to the student through the web interface.

Project Folder Structure

File / Folder	Purpose
app.py	Main Flask web application — entry point
agent.py	Module 1: Intelligent Agent design and PEAS model
search.py	Module 2: BFS and A* search algorithm implementations
knowledge_base.json	Module 3: All FOL facts and rules in structured JSON format
inference.py	Module 3: Forward and Backward Chaining inference engine
planner.py	Module 4: Classical and Hierarchical response planning
learner.py	Module 5: Decision Tree and Reinforcement Learning engine
expert_rules.py	Module 5: Expert System production rules
templates/index.html	Web-based chat interface for student interaction
data/training_data.csv	Historical Q&A pairs for Decision Tree training

7. Project Timeline

Phase	Duration	Title	Deliverables
Phase 1	Week 1–2	Research & Design	System design document, KB schema, architecture diagram
Phase 2	Week 3–4	Core AI Development	Working search engine, FOL rules, agent interface in Python

Phase	Duration	Title	Deliverables
Phase 3	Week 5–6	Learning & Expert System	Decision Tree classifier, RL module, Expert System rules
Phase 4	Week 7	UI & Integration	Web interface, end-to-end integration, user testing
Phase 5	Week 8	Testing & Documentation	Test report, final project report, presentation

8. Expected Outcomes

- **A fully functional Intelligent College Assistant Chatbot deployable on a college website or WhatsApp.**
- Demonstration of all 8 core AI algorithms including BFS, A*, Forward/Backward Chaining, Decision Tree, RL, CSP, and Expert System rules.
- A structured knowledge base of college information represented in First-Order Logic.
- A web-based chat interface accessible from desktop and mobile browsers.
- A comprehensive project report with chapters aligned to all 6 syllabus modules.
- A chatbot capable of handling at least 50 common student queries accurately.
- A learning-enabled system that improves its accuracy based on student feedback over time.

9. Relevance & Justification

Academic Relevance

This project covers all six modules of the S5 Artificial Intelligence syllabus in an integrated manner. Each AI technique implemented directly corresponds to a module topic, making every chapter of the project report academically substantive and easy to examine.

Industry Relevance

Chatbot development is among the most sought-after skills in the software industry today. By implementing NLP, search algorithms, logical inference, and machine learning in a single project, students gain hands-on experience with the full AI stack used in industry applications at companies ranging from banks and hospitals to e-commerce and education platforms.

Practical Relevance

The completed chatbot can be directly deployed within the college to serve real students, reducing the workload on administrative staff and providing 24/7 instant support. This makes the project both academically rigorous and practically impactful.

Research Relevance

The project combines symbolic AI (rules, logic, expert systems) with statistical AI (machine learning, reinforcement learning), a combination known in current research as Neuro-Symbolic AI. This positions the project at the frontier of modern AI research while remaining grounded in the classical techniques of the curriculum.

10. Technology Stack

Category	Technology	Purpose
Backend	Python 3.x + Flask	Core AI logic and web server
NLP	NLTK / spaCy	Tokenisation and query understanding
Machine Learning	scikit-learn	Decision Tree classifier training
Knowledge Base	JSON / SQLite	Storage of FOL facts and rules
Frontend	HTML5 + CSS3	Student-facing chat interface
Deployment	Heroku / Local Server	Hosting the chatbot application

11. Conclusion

The Intelligent College Assistant Chatbot is a comprehensive, multi-faceted AI project that integrates search algorithms, knowledge representation, logical inference, planning, and machine learning into a single, practical application. It addresses a real problem faced by students and colleges while demonstrating mastery of the full S5 Artificial Intelligence syllabus. The project is feasible within the academic timeline, implementable using freely available Python tools, and capable of producing measurable real-world impact through actual deployment in a college environment.

We respectfully request supervisor approval to proceed with this project.

Submitted by: _____

Date: _____